Reproducing conditional DiffTraj on simulated vehicle trajectories

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Experimental question

Can the published DiffTraj training and sampling procedure generate vehicle trajectories with the same spatial and distance distributions as a supplied traffic simulation? I first trained the released architecture on the supplied vehicle records. I then repeated the experiment on two nested sets of complete taxi journeys generated with SUMO. Each training set has two independently trained models. A paired optimizer repeat uses the same 20,000 taxi trips and the same seeds.

What was reproduced

Zhu et al. train a diffusion model to generate a sequence of locations from a trip description. Their main experiments use taxi GPS from Chengdu and Xi'an. I verified the authors' public Chengdu checkpoint and request file by drawing 500 trajectories (141,747 output positions), then retrained the same conditional network from scratch on the SUMO coordinates. The checkpoint exercise verifies sampling. Chengdu samples. The original observed Chengdu and Xi'an evaluation trips are needed to independently reproduce the numerical scores of Table 1. On the new 10,000-taxi training set, both seeds recovered a trip-length distribution close to the observed one. On 20,000 trips, an optimizer change recovered an unstable repeat: the two corrected seeds both generated a median journey of 5.08 km against an observed 5.59 km. Reproducing the joint origin–destination frequencies proved harder. DiffTraj, Sections 4–5 and Appendices A–C. Author release.

Why generate a second dataset?

The supplied export yielded 1,478 training vehicle tracks, including only 69 identified as taxi or ride-hailing. Splitting those tracks into shorter windows would reuse the same journeys and change their endpoints. In a discussion about how many trips survive preprocessing, the repository owner suggested checking what a lower point cutoff would retain and described training collections on the order of 100,000 trajectories as useful in practice. This motivated generating additional independent trips while keeping the original point-count filter. I chose a balanced five-class archive so the same simulations can support a later vehicle-class experiment. Here only its taxi subset enters training, matching the vehicle type used in DiffTraj's main study.

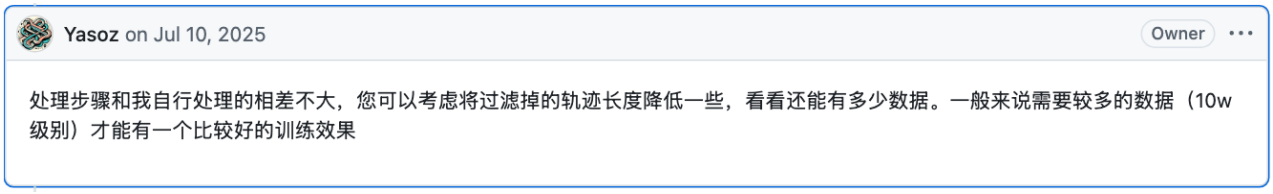


Figure 1. The repository owner's original Chinese reply, 10 July 2025. The scale mentioned is approximately 100,000 trajectories in total. The comment gives an order of magnitude, with no per-class target. Open the original GitHub comment.

Sampling from the authors' Chengdu checkpoint

The public release supplies trained model weights and request attributes. I first loaded those files and sampled 500 Chengdu trips with seed 42 using the released demonstration. Restoring each trajectory to its requested point count gave 141,747 positions. The median trip had 259 positions. This is a direct run of the published sampling example before any training on SUMO.

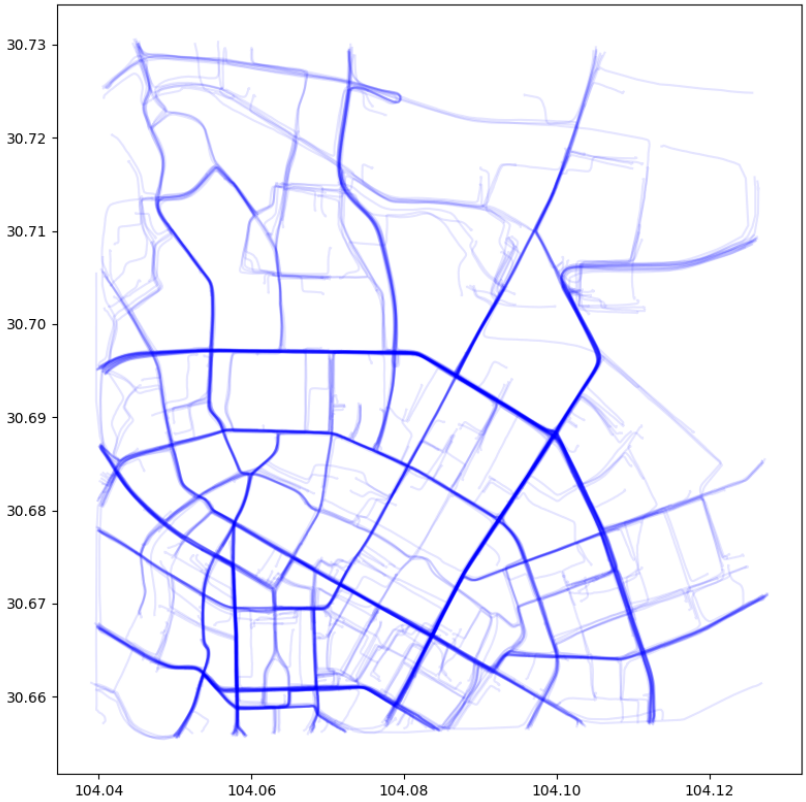


Figure 2. Five hundred generated Chengdu taxi trajectories from the authors' weights, overlaid in longitude (horizontal) and latitude (vertical). Darker lines indicate repeated paths. Archived run summary.

The sample concentrates along several recurring corridors. Calculating the four distributional errors also requires the corresponding observed GPS sequences. The public request vectors describe each trip in aggregate and cannot supply its observed point density or full path. For the SUMO retraining experiments, both the observed reference coordinates and the generated coordinates are available, so all four measures and the paired maps can be reported.

Data and preprocessing

Supplied vehicle observations

The input for this experiment was `midimost_with_class2.csv`. Each row contains simulation time, vehicle ID, planar position, simulator speed and a class label. I removed pedestrians, grouped by vehicle ID, sorted by time and collapsed duplicate timestamps. Each retained trajectory needed at least 120 original observations, finite coordinates, movement and no gap over two seconds. Of 1,896 vehicle IDs, 43 fell below the point cutoff. The remaining 1,853 tracks contain private cars, motorcycles, bicycles, buses and taxis or ride-hailing vehicles. A vehicle-ID split gave 1,478 training, 198 development and 177 reserved tracks. These tracks span the portions of a vehicle's journey visible in the export. The separate passenger-only file `vehicle_positions_TS_New.csv` was outside this experiment.

New SUMO journeys

I ran independent traffic episodes on the public Monaco SUMO Traffic (MoST) road network. For taxis, passenger cars, motorcycles and bicycles, the simulation placed origin–destination requests on roads accessible to each class. Two road edges were sampled with a straight-line separation of 1.1–6.5 km, and departure was drawn within the first 900 simulation seconds. SUMO routed the vehicle between those edges in traffic. Buses used the scenario's lines and stops with synthetic departures. I recorded positions every second, checked departure and arrival against SUMO's trip records, and retained continuous completed journeys with at least 120 positions, at least 100 m of movement and no one-second position change above 65 m. The simulations disabled teleporting. I checked for duplicate interpolated paths and assigned complete episodes to disjoint training, development and test partitions. MoST scenario.

Collection	Training	Development	Reserved/test
Supplied observations, all vehicle classes	1,478	198	177
First simulation archive, five classes	50,000	5,000	5,000
Expanded simulation archive, five classes	100,000	5,000	5,000
Taxi subset used in first / expanded fit	10,000 / 20,000	1,000	1,000

The five classes have equal training counts. The expanded training set includes the first release. The two taxi fits use identical development requests. The evaluation on the supplied set uses all 177 reserved IDs.

Common model inputs

For each track, I computed native point count, elapsed seconds and cumulative Euclidean distance from the original positions in metres. The supplied median sampling interval is 0.25 s. Thus 120 regularly spaced points span 29.75 s in that export and 119 s in the one-second simulations. I then interpolated each track linearly along sample index to 200 two-dimensional positions, and fitted coordinate standardization and continuous-feature standardization on the training split. Scoring restores each generated trip to its requested number of observations. Origin and destination cells use the training coordinate range on a 16×16 grid.

DiffTraj training and sampling

A trip request has eight attributes. The three categorical attributes are the five-minute departure bin and the start and end cells. The five continuous attributes are observed travel distance L, duration T, native point count n, average interpoint displacement $L/(n-1)$ and average speed L/T. Distances come from the observed coordinate sequence. For newly simulated trips, the departure bin uses the actual simulation clock at departure, binned in five-minute intervals. Both datasets use only positions as the output channels. The supplied time-varying speed and vehicle-class labels do not enter this baseline's loss or request vector.

For each training example, x_0 is the standardized 2×200 sequence and c is its eight-attribute request. At a sampled diffusion step t, draw Gaussian noise ϵ of the same shape. With a linear variance schedule β_t and $\bar{\alpha}_t = \prod_{s \le t}(1 - \beta_s)$, the training input and loss are

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1 - \bar{\alpha}_t}\, \epsilon, \qquad \mathcal{L} = \mathrm{E}_{x_0, c, t, \epsilon}\left[\frac{1}{400} \left\| \epsilon - \hat{\epsilon}_\theta(x_t, t, c) \right\|_F^2\right].$$

The temporal U-Net learns to predict the injected noise at all 400 coordinate entries. In the released implementation two calls to the shared U-Net use the encoded request and a separately encoded all-zero request. The combined prediction is $\hat{\epsilon}_\theta = \epsilon_c + 3(\epsilon_c - \epsilon_\emptyset)$. To generate a trip, start from Gaussian noise, hold c fixed, and repeatedly use this prediction to recover coordinates. Each request receives one draw for each trained seed.

Training set	Batch	Epochs	Updates	L20s per seed
Supplied vehicles	32	200	9,400	2
Simulated taxis, 10,000	220	200	9,200	1
Simulated taxis, 20,000	440	200	9,200	1

The six original fits use seeds 42 and 43, AdamW at a fixed learning rate of 2×10^{-4}, 500 diffusion noise levels (β from 0.0001 to 0.05) and 100 DDIM reverse updates with stride 5 and $\eta = 0$. The final epoch's ordinary weights are scored. The runs share the denoising objective, conditioning architecture and sampling schedule. The three batch sizes give comparable counts of optimizer updates. The 20,000-trip run sees twice as many examples per epoch as the 10,000-trip run and its gradient estimates use larger batches.

Implementation choices to reproduce a run

The released denoiser is imported directly. Local code adapts input units, vehicle or episode splits, feature normalization, batching and evaluator boundaries to the datasets. Sampling runs in evaluation mode with dropout disabled. The public demonstration notebook leaves its model in training mode. Coordinates are planar metres. A separate alignment check supports using MoST as a reference map for the supplied export. Released U-Net and guidance. Released configuration.

What the denoising loss shows

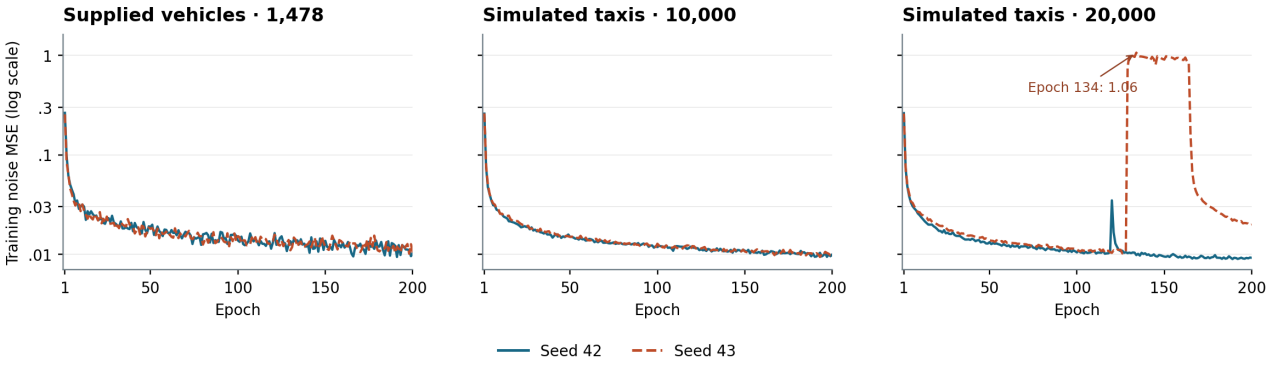


Figure 3. Training noise-prediction MSE by epoch for all six fits. The value is averaged over the first training rank in the original logs. The supplied runs used two ranks; the taxi runs used one. The objective measures noise prediction on perturbed training tracks. The maps on the next pages use full reverse sampling from noise.

At 10,000 taxi trips, both runs ended with a noise MSE of approximately 0.010 and produced similar distance distributions. At 20,000 trips, seed 43 rose from 0.0102 at epoch 126 to 1.063 at epoch 134. It returned to 0.0199 by epoch 200. Seed 42 ended at 0.0092. The change in seed 43 also appeared in the generated trips: the median rose to 24.97 km, compared with 5.59 km in the observations for those same requests. The loss spike dates the instability in that fit. The logs do not distinguish its possible causes.

How to read the maps

Each column follows one observed request and compares its coordinate sequence with one draw. The two rows are independent training seeds. A model knows the cells, time and aggregate trip attributes; several paths can satisfy the same request. The overlays show how the generated path sits relative to the observed one, including the locations of its turns and endpoints. The city map provides a common spatial reference for the examples.

For the supplied tracks, the original network file was unavailable. A coordinate check against the public MoST map placed all 4,841 sampled positions within 5 m of a lane centreline without shifting coordinates. The new training data were simulated on the MoST network used in their maps. Map alignment record.

Generated trajectories on the supplied data

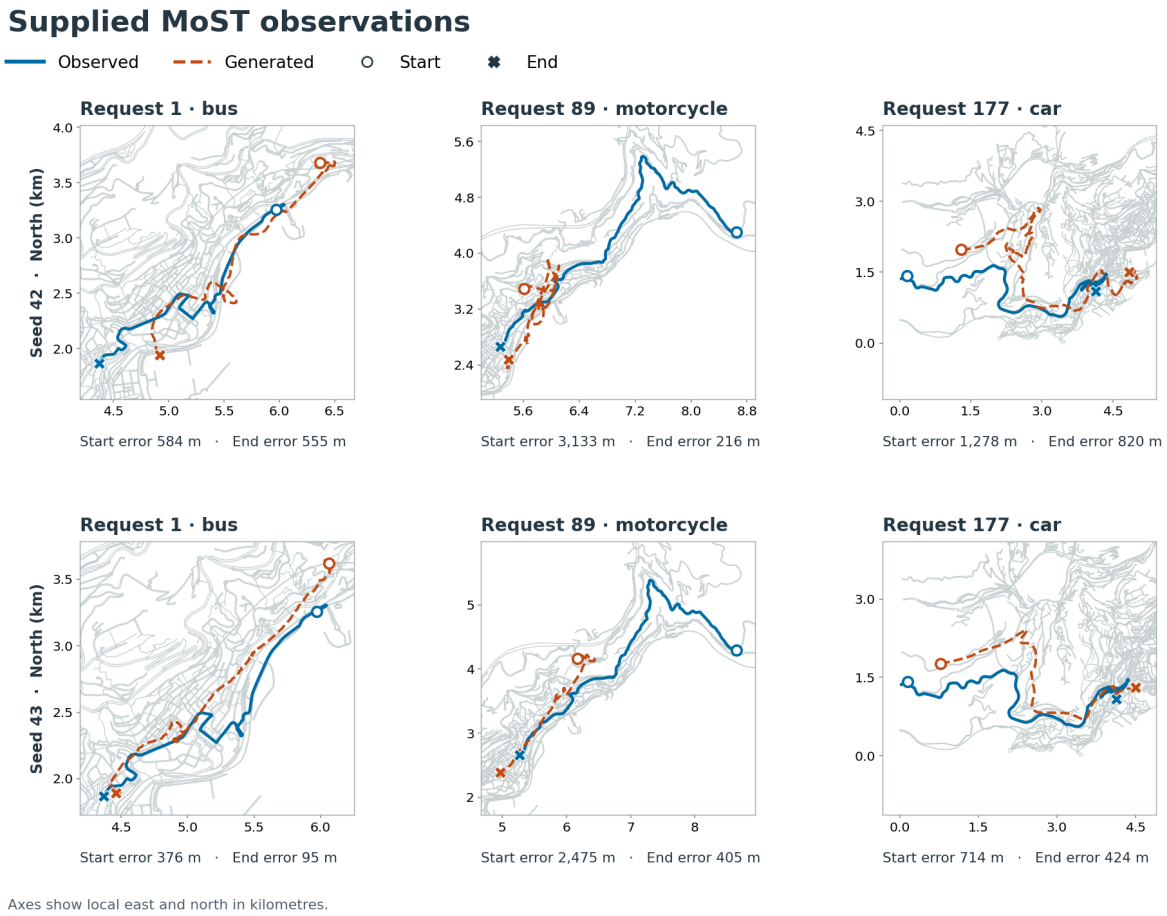


Figure 4. Supplied simulation, small-batch training. Each column overlays the observed vehicle coordinates and the generated coordinates for the same held-out request. Requests 1, 89 and 177 are a bus, a motorcycle and a private car. These are the first, middle and last entries in saved order. Marker pairs show the departure and arrival offsets in metres.

What the full set shows

In the middle column, both seeds place the motorcycle destination within 405 m of its observation, but the origins are 3.13 km and 2.48 km away. The bus in the first column ends within 95 m for seed 43. The 177 observed tracks had a median length of 5.86 km. The generated medians were 10.52 km for seed 42 and 5.66 km for seed 43. Across the full distributions, the corresponding length Wasserstein distances were 6.97 and 0.49 km. This distance measures the average displacement between corresponding quantiles of the two length distributions. A median alone would conceal the much broader spread of one run.

Generated trajectories from 10,000 taxi journeys

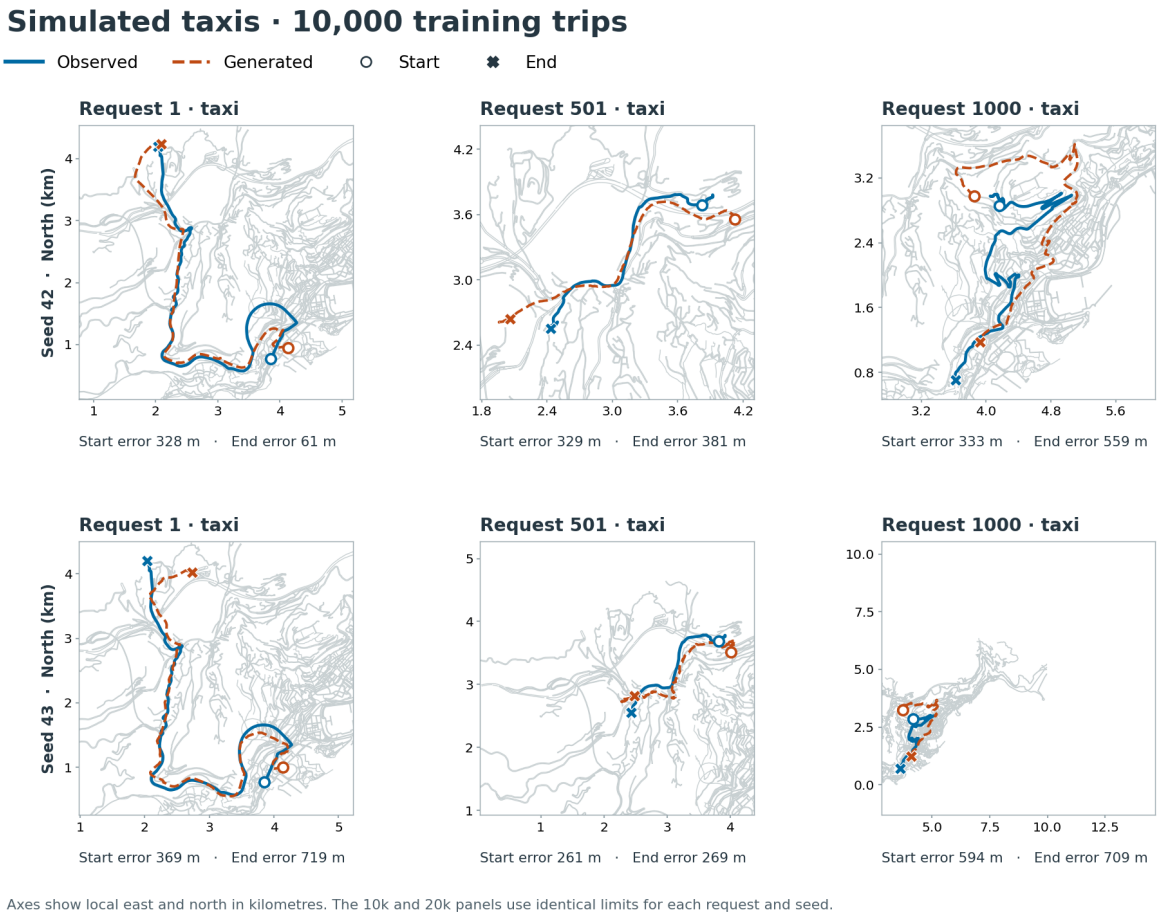


Figure 5. Taxi draws from the 10,000-trip training subset. The three requests come from the saved development set, in fixed order. Figure 6 repeats these same requests after expanding the training data, so a change in shape can be examined for the same physical request and sampling noise, at the same map scale. Each model uses its own training normalization.

For request 501, the observed journey is 2.82 km. Seed 42 starts 329 m from its observed origin and finishes 381 m from the observed destination. The 1,000 observed journeys have a median length of 5.59 km. Length Wasserstein distances are 0.64 and 0.66 km over the two seeds. Start-position median errors are 295 and 319 m; end-position median errors are 321 and 309 m. The evaluation grid cells are roughly 623 m wide and 393 m high. At this resolution, both ends fall in the requested cells for 24.7% and 22.6% of draws. A discrepancy of a few hundred metres can cross a cell boundary.

Generated trajectories from 20,000 taxi journeys

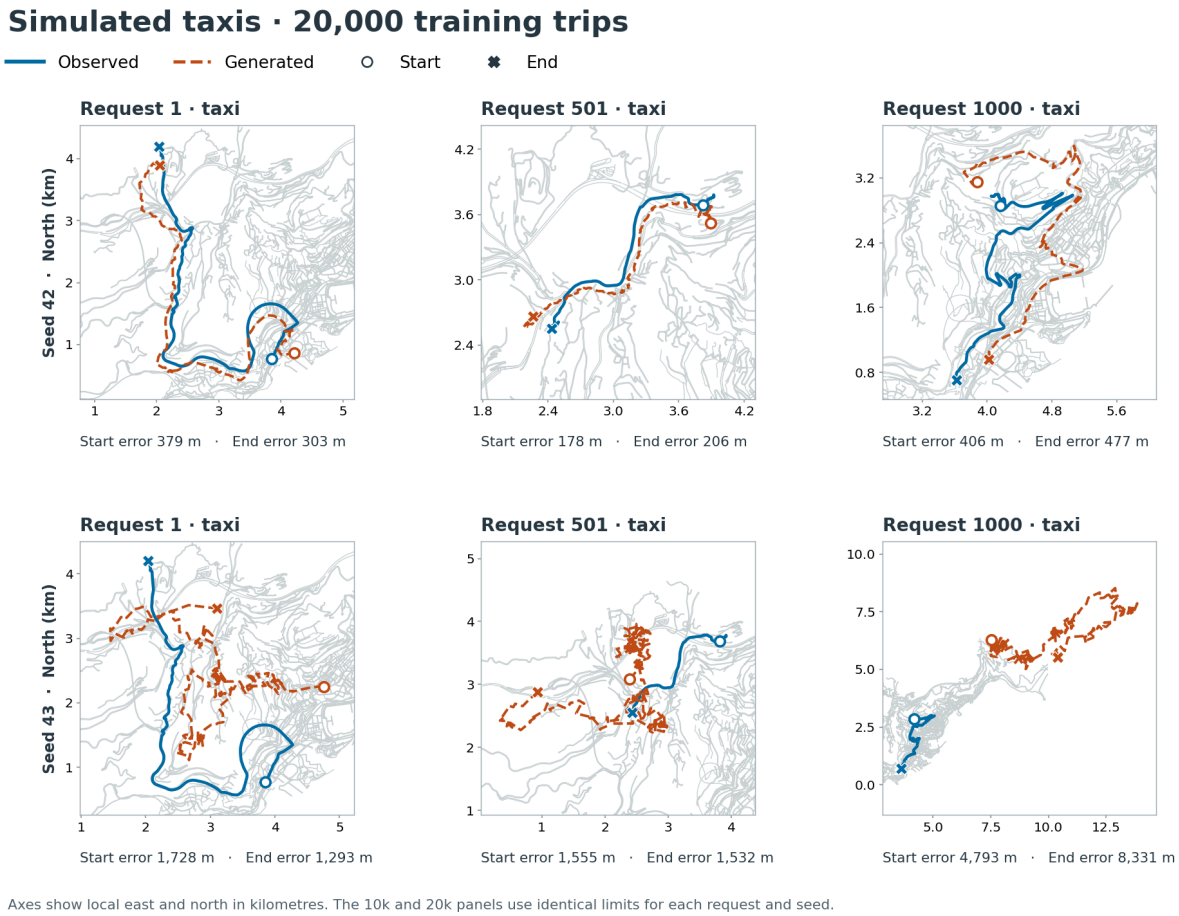


Figure 6. Taxi draws from the expanded 20,000-trip training subset. The two seeds and the request order correspond to Figure 5. Both fits are included in the quantitative comparison.

For the same request 501, the seed-42 errors fall to 178 m at departure and 206 m at arrival. Seed 43 instead misses both endpoints by about 1.5 km. Across all 1,000 requests, seed 42 generated a median of 5.91 km. Its length Wasserstein distance fell to 0.36 km. Seed 43 produced a median of 24.97 km and a length Wasserstein distance of 32.62 km. The percentage of requests with both endpoints in the requested cells was 23.6% for seed 42 and 0.1% for seed 43. On these identical development requests the original optimization schedule gave very different results across seeds.

Where the complete evaluation sets put their points

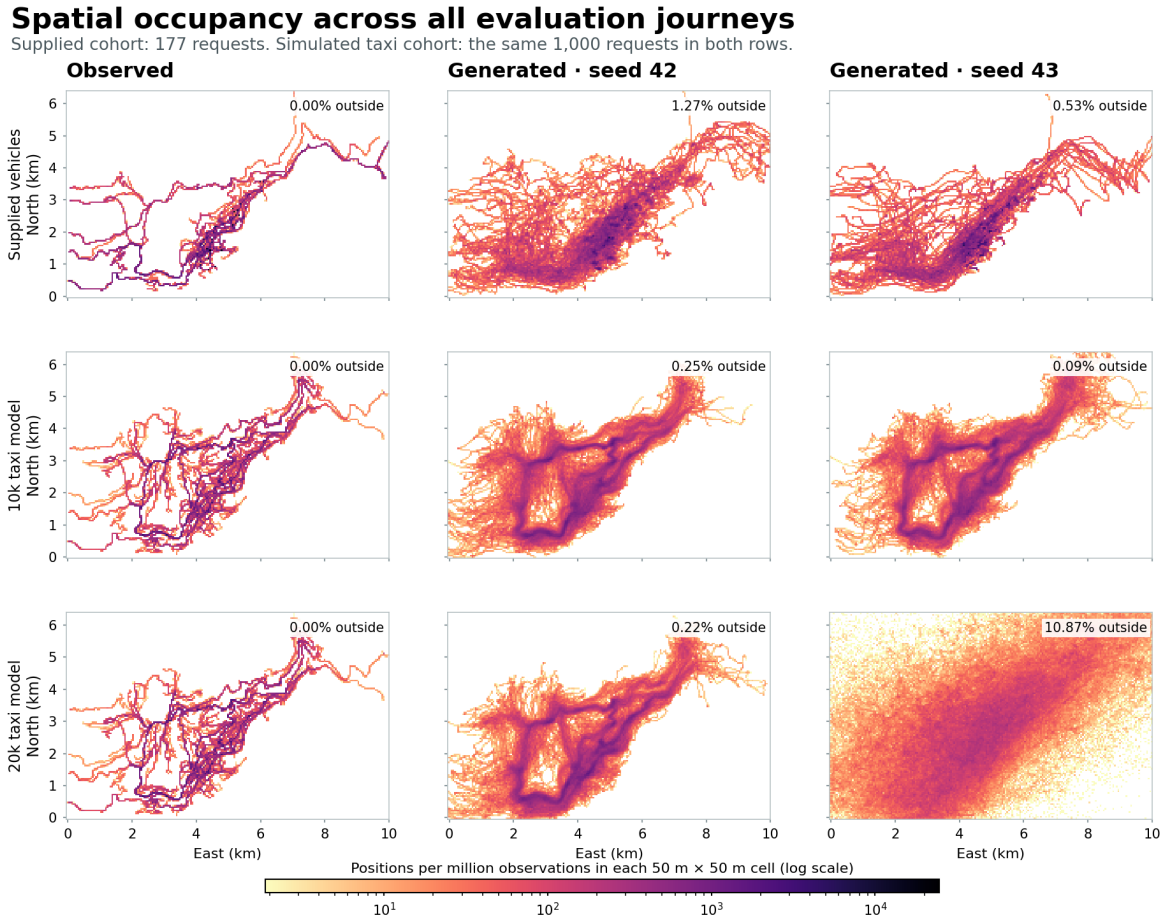


Figure 7. Observed and generated point distributions for all scored requests, following the geographic comparison in DiffTraj Figure 2. Every panel uses the same city extent, 50 m cells and logarithmic colour scale. The two observed taxi panels use the same trips. The printed outside fraction includes all positions beyond the displayed extent.

The observed points concentrate along narrow corridors. Both 10,000-taxi samples reproduce the main concentration of trips but spread positions over more of the map. For the same 532,033 positions per panel, the observations occupy 4,559 cells, while seeds 42 and 43 occupy 9,007 and 9,621. With 20,000 training trips, seed 43 puts 10.87% of positions outside the map window, compared with 0.22% for seed 42. These patterns explain why this run also deserves inspection at the level of individual requests.

The 50 m cells resolve the spatial patterns in the figure. The numerical density error in the score tables uses the 16×16 grid specified for scoring. Published geographic comparison.

Repeat training on the same 20,000 taxi journeys

The original seed-43 loss rose sharply at epoch 134. To test whether it could recover, I reran both seeds from initialization on the same training file with the same network, loss, batch of 440 and 9,200 updates. Before each AdamW update I clipped the global gradient norm at 1.0. The learning rate followed a cosine schedule from 2×10^{-4} to 2×10^{-5} over 200 epochs. These two changes form one intervention.

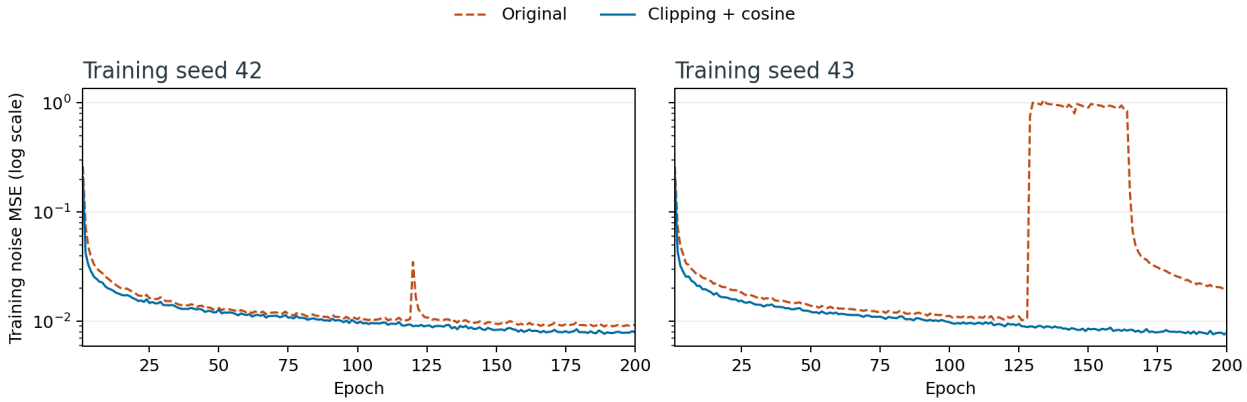


Figure 8. Training noise MSE before and after the optimizer change. Both corrected runs remained below 0.010 over epochs 100–200. The final MSE was 0.00793 for seed 42 and 0.00786 for seed 43.

Comparison at a fixed training budget

The primary rows compare the final weights at epoch 200. Every row below uses the same 1,000 development requests, initial sampling noise, grid and distance bins. I also saved the weights with the smallest denoising MSE on 512 fixed development trips with fixed noise. That subset overlaps the evaluation population, so the selected checkpoint is an additional diagnostic.

Seed	Weights	Density JS	OD JS	Length JS	F1	Length W1 (km)
42	Original	0.0211	0.4031	0.0101	0.900	0.36
42	Clipped + cosine	0.0038	0.2466	0.0124	0.900	0.46
42	Val. min., epoch 200	0.0038	0.2466	0.0124	0.900	0.46
43	Original	0.1531	0.6374	0.3499	0.500	32.62
43	Clipped + cosine	0.0029	0.2285	0.0118	0.950	0.48
43	Val. min., epoch 197	0.0026	0.2340	0.0136	0.900	0.44

Seed 43 recovered on all four distributional metrics. Its trip-length Wasserstein distance dropped from 32.62 to 0.48 km. For seed 42, density and OD improved and hotspot F1 stayed at 0.90. Its length Wasserstein distance rose modestly from 0.36 to 0.46 km. Both corrected runs generated a median trip length of 5.08 km, against an observed 5.59 km.

Paired draws after the optimizer change

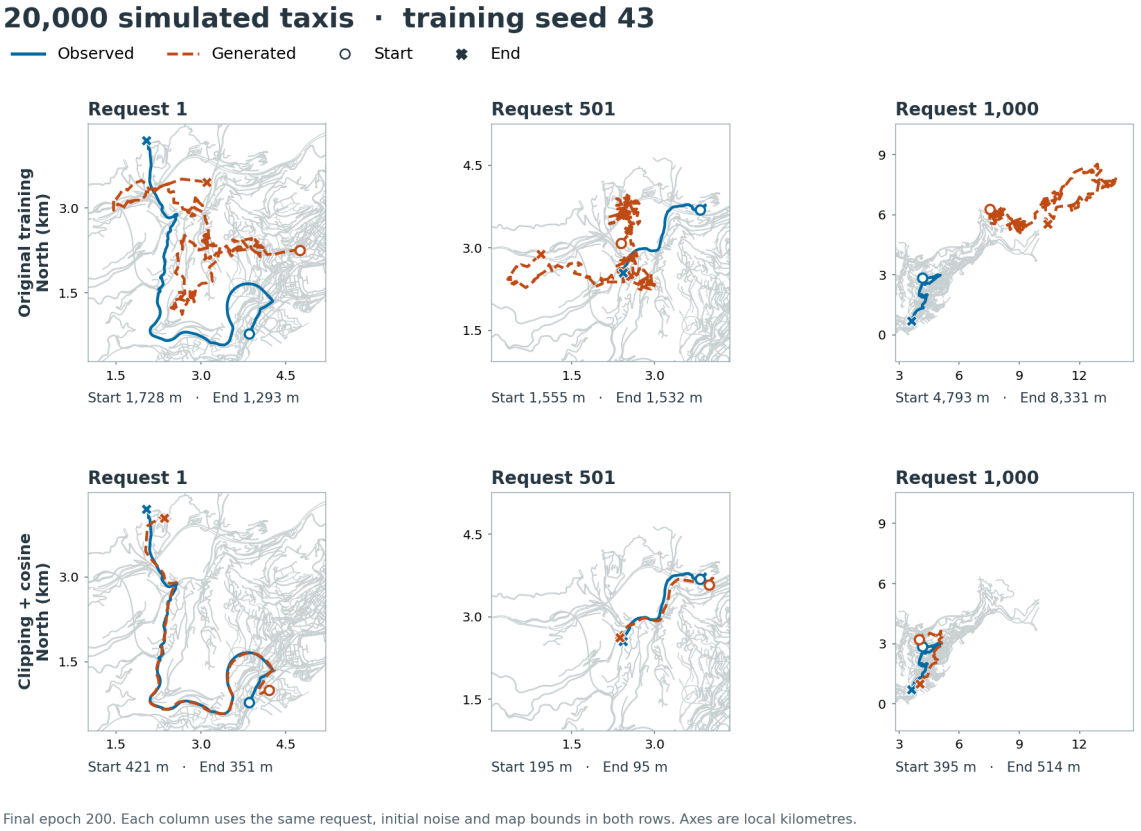
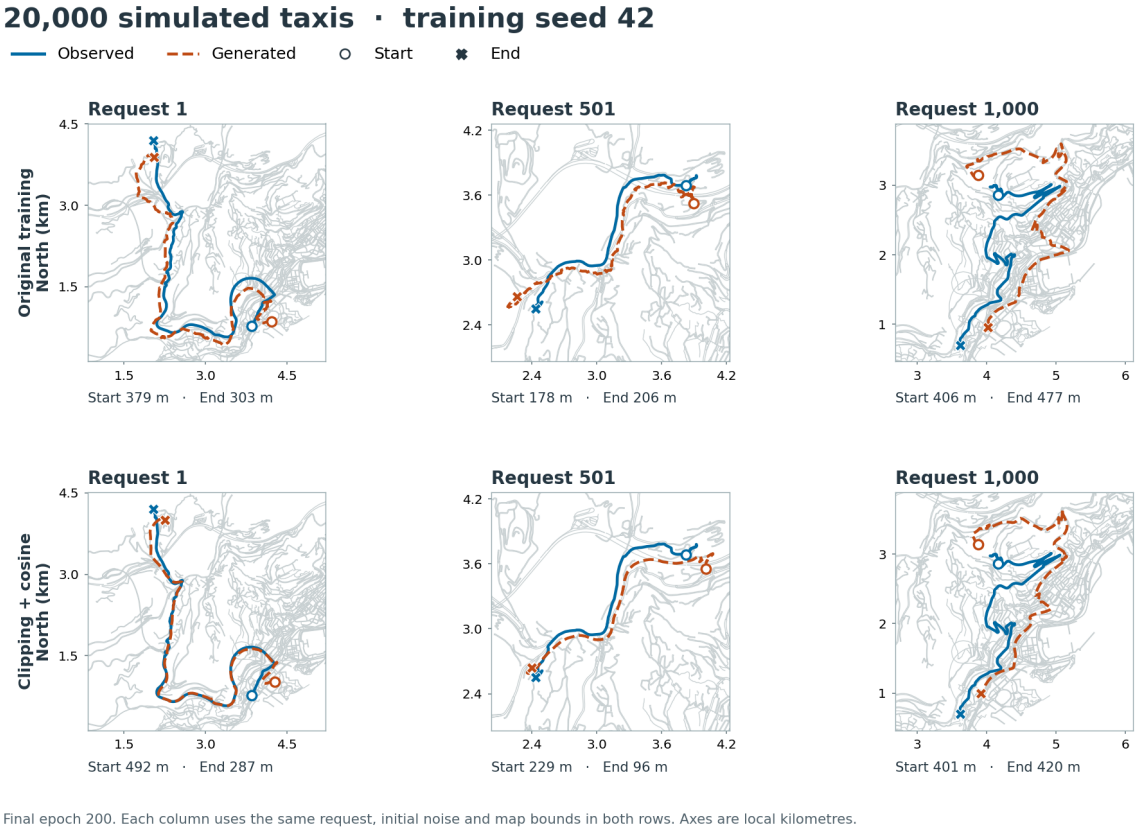


Figure 9. Seed 43 at epoch 200. The top row uses original training and the bottom row uses clipping and cosine decay. Each column uses identical observed positions, sampling noise and axis limits. The columns are requests 1, 501 and 1,000 in the original evaluation order.

For request 501 the original sample started 1,555 m and ended 1,532 m from the observed positions. After retraining those discrepancies fell to 195 m and 95 m. Across all 1,000 requests, median start and end discrepancies fell from 1,542 and 1,710 m to 225 and 213 m. Both endpoints landed in their requested cells on 563 trips, up from one trip. The model was given the endpoint cells and trip summaries; exact endpoint coordinates only enter this comparison after generation.

Effect on the previously better-performing seed



Final epoch 200. Each column uses the same request, initial noise and map bounds in both rows. Axes are local kilometres.

Figure 10. Seed 42 at epoch 200, using the same three requests as Figure 9. Within each column, the original and corrected draws share the same map extent.

For request 501 the end discrepancy fell from 206 to 96 m, while the start discrepancy rose from 178 to 229 m. Over all requests the median start and end discrepancies fell from 315 and 301 m to 232 and 218 m. Joint endpoint-cell hits rose from 236 to 545 out of 1,000. The small deterioration in seed 42's distance distribution coexists with better aggregate endpoint placement.

City-wide distribution after retraining

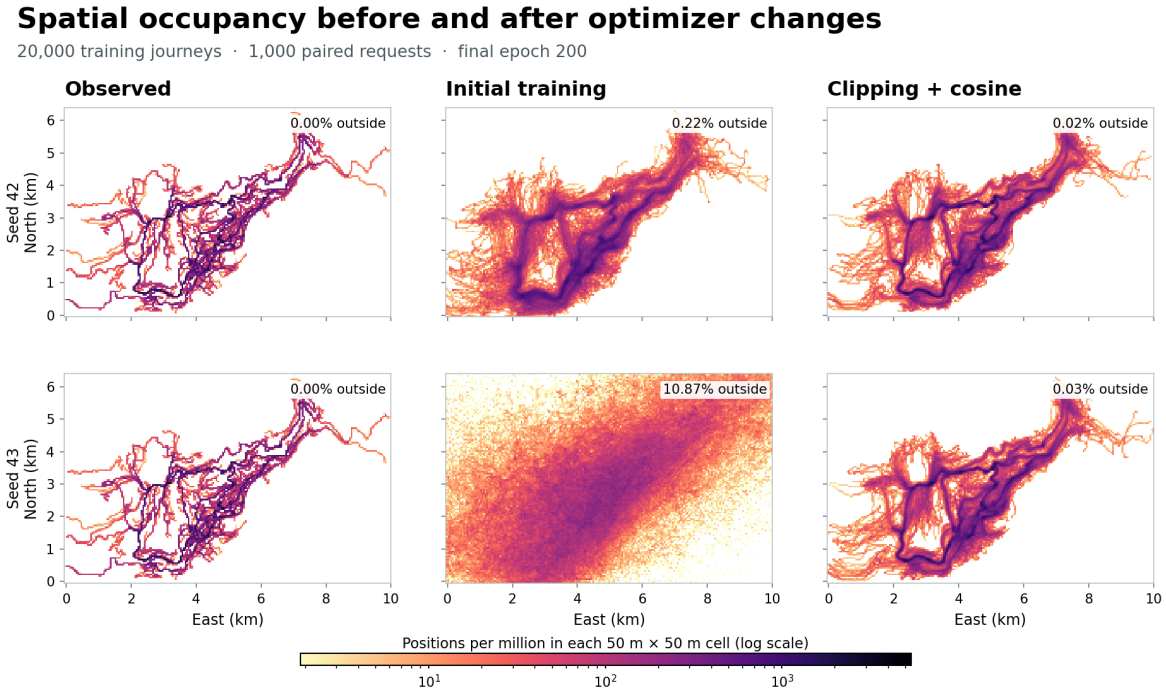


Figure 11. All 1,000 development requests, with 532,033 evaluated positions in each panel. The two rows show the two training seeds. All panels use the same 50 m cells, map extent and colour scale. Each panel's denominator includes points outside the displayed map.

For seed 43, positions beyond the map window fell from 57,807 to 167 (10.87% to 0.031%). For seed 42 they fell from 1,178 to 95 (0.221% to 0.018%). This check preserves every generated point, including positions that the example maps cannot show at city scale. The two corrected runs now give similar city-wide distributions and endpoint errors on the shared requests.

Comparison with the published experiment

Every local row scores one generated draw for each observed request. **Density JS** compares the distribution of all points over cells. **OD JS** compares the frequencies of ordered start–end cell pairs. **Length JS** compares histograms of cumulative trip distance. These three are Jensen–Shannon divergences using natural logarithms locally. Lower values indicate closer distributions. **Pattern F1** measures overlap of the most frequently visited cells. A larger value indicates greater overlap. The paper leaves the hotspot count unspecified. The local calculation uses the 20 most frequent cells and 20 training-quantile distance bins. The taxi size comparison uses one training grid and one set of bins from the 10,000-trip run for both sizes.

Source and seed	Density JS	OD JS	Length JS	Pattern F1
Published Chengdu (Table 1)	.0055	.0154	.0169	.823
Published Xi'an (Table 1)	.0126	.0165	.0203	.764
Published Porto (Table 5)	.0087	.0132	.0242	.847
Supplied vehicles, 42	.1110	.5597	.2058	.650
Supplied vehicles, 43	.0690	.5564	.0536	.800
Simulated taxis 10,000, 42	.0191	.4166	.0193	.750
Simulated taxis 10,000, 43	.0130	.4156	.0237	.800
Simulated taxis 20,000, 42	.0211	.4031	.0101	.900
Simulated taxis 20,000, 43	.1531	.6374	.3499	.500
20,000 taxis, corrected, 42	.0038	.2466	.0124	.900
20,000 taxis, corrected, 43	.0029	.2285	.0118	.950

Both 10,000-taxi fits produced length JS near the published main-city values (.0169–.0203), and hotspot overlap in a similar range. The corrected 20,000-taxi runs give lower density and length JS and higher hotspot F1 than the reported values in both main cities. Joint OD error remains the largest gap, at .2466 and .2285 versus .0154 and .0165. These reference values come from GPS observations in different cities.

Matching the published evaluation count

For a separate check, the frozen 10,000-taxi models generated one sample for each of 3,000 journeys from 44 fresh episodes. At the paper's main evaluation count, density / OD / length JS and hotspot F1 were .0204 / .3154 / .0175 / .750 for seed 42 and .0114 / .3006 / .0212 / .850 for seed 43. OD error remains well above the published values on this independent sample. The cohort and sample count both change relative to the 1,000-trip development table.

Why the results differ

Demand and sample size. The paper lists roughly 3.49 million Chengdu trips and 2.18 million Xi'an trips and describes evaluation samples of 3,000 for its main experiments. The supplied evaluation has 177 mixed-vehicle requests. Comparing 177 observed development trips against 177 observed reserved trips already gives OD JS .5144. In the expanded experiment the 1,000 observed taxi trips occupy 817 OD pairs. The most common pair occurs only eight times. Sparse support makes the empirical joint histogram sensitive to endpoint shifts and sample composition. Independent observed samples provide context. Generating from the same requests is a different comparison.

Training variability and measurement. The 20,000-trip loss spike coincides with poorer generated distances and endpoints. The supplied source also mixes vehicle classes and uses a finer observation clock. Changing from that source to the new taxi set changes both population and demand. The paired 10,000 versus 20,000 experiment is the closer test of expansion. City bounds, distance bins and hotspot counts affect the comparison with Table 1. After retraining, both 20,000-taxi seeds give similar trip lengths. Joint OD frequencies remain the largest discrepancy.